The University of Texas at Austin
In-Class Work 10

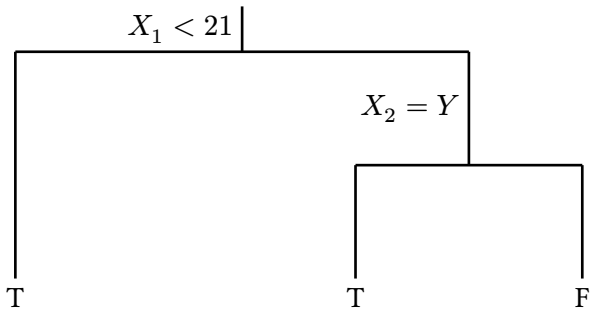
M339G Predictive Analytics

November 11, 2026

Classification Trees

Actuarial Exam Problems.

Problem 10.1. *Source: Exam MAS-II, Fall 2018.*
You are given the following classification decision tree and data set:



i	X_1	X_2	Y
1	12	Y	T
2	23	N	F
3	4	Y	F
4	32	Y	F
5	22	N	T
6	30	Y	T
7	18	N	T

Determine the relationship between the classification error rate, the Gini index, and the cross-entropy, summed across all nodes.

a. cross-entropy $>$ Gini index $>$ classification error rate
b. cross-entropy $>$ Gini index $=$ classification error rate
c. classification error rate $>$ Gini index $>$ cross-entropy
d. Gini index $>$ cross-entropy $>$ classification error rate
e. The correct answer is not given by a., b., c., or d.

Solution. The correct answer is **a.**
Sort the observations into the terminal nodes:

- Left node ($X_1 < 21$): observations $1, 3, 7$ with responses T, F, T. So $\hat{p}_T = \frac{2}{3}, \hat{p}_F = \frac{1}{3}$.

- Middle node ($X_1 \geq 21$, $X_2 = Y$): observations $4, 6$ with responses F, T. So $\hat{p}_T = \hat{p}_F = \frac{1}{2}$.
- Right node ($X_1 \geq 21$, $X_2 = N$): observations $2, 5$ with responses F, T. So $\hat{p}_T = \hat{p}_F = \frac{1}{2}$.

Classification error rate, $E = 1 - \max_k \hat{p}_k$:

$$\frac{1}{3} + \frac{1}{2} + \frac{1}{2} = \frac{4}{3} \approx 1.333.$$

Gini index, $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$:

$$2 \cdot \frac{2}{3} \cdot \frac{1}{3} + 2 \cdot \frac{1}{2} \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{4}{9} + 1 \approx 1.444.$$

Cross-entropy, $D = -\sum_k \hat{p}_k \ln \hat{p}_k$:

$$-\left(\frac{2}{3} \ln \frac{2}{3} + \frac{1}{3} \ln \frac{1}{3}\right) + 2 \ln 2 \approx 0.6365 + 1.3863 = 2.0228.$$

Therefore, cross-entropy $>$ Gini index $>$ classification error rate.

Problem 10.2. *Sample SRM Problems.*
A classification tree is being constructed to predict if an insurance policy will lapse. A random sample of 100 policies contains 30 that lapsed. You are considering two splits:

- Split 1: One node has 20 observations with 12 lapses and one node has 80 observations with 18 lapses.
- Split 2: One node has 10 observations with 8 lapses and one node has 90 observations with 22 lapses.

The total Gini index after a split is the weighted average of the Gini index at each node, with the weights proportional to the number of observations in each node. The total entropy after a split is the weighted average of the entropy at each node, with the weights proportional to the number of observations in each node.

Determine which of the following statements is/are true.

I. Split 1 is preferred based on the total Gini index.
II. Split 1 is preferred based on the total entropy.
III. Split 1 is preferred based on having fewer classification errors.

a. I only
b. II only
c. III only
d. I, II, and III
e. The correct answer is not given by a., b., c., or d.

Solution. The correct answer is **e.**
Smaller values of the total Gini index, total entropy, and number of classification errors are preferred. The lapse proportions in the nodes are 0.6 and 0.225 for Split 1, and 0.8 and $\frac{22}{90} \approx 0.2444$ for Split 2.

- I is false. Using $G = 2\hat{p}(1-\hat{p})$ at each node:
  - Split 1: $0.2 \cdot 2(0.6)(0.4) + 0.8 \cdot 2(0.225)(0.775) = 0.096 + 0.279 = 0.375$.
  - Split 2: $0.1 \cdot 2(0.8)(0.2) + 0.9 \cdot 2(0.2444)(0.7556) \approx 0.032 + 0.3324 = 0.3644$.

  Split 2 has the smaller total Gini index.
- II is false. Using $D = -\hat{p}\ln\hat{p} - (1-\hat{p})\ln(1-\hat{p})$ at each node:
  - Split 1: $0.2(0.6730) + 0.8(0.5332) \approx 0.1346 + 0.4265 = 0.5611$.
  - Split 2: $0.1(0.5004) + 0.9(0.5561) \approx 0.0500 + 0.5005 = 0.5506$.

  Split 2 has the smaller total entropy. (The comparison is the same for any base of the logarithm.)
- III is false. Each node predicts its majority class.
  - Split 1: 8 errors in the first node and 18 in the second, for 26 errors.
  - Split 2: 2 errors in the first node and 22 in the second, for 24 errors.

  Split 2 has fewer classification errors.

Conceptual Problems.

Problem 10.3. Determine which of the following statements concerning the splitting criteria for classification trees is/are true.

I. The Gini index and the cross-entropy are more sensitive to node purity than the classification error rate.
II. A split that produces two terminal nodes with the same predicted class can never be useful.
III. The Gini index of a node is small when every class proportion in that node is close to 0 or 1.

a. I only
b. II only
c. I and III only
d. II and III only
e. The correct answer is not given by a., b., c., or d.

Solution. The correct answer is **c.**

- I is true: the classification error rate depends only on the largest class proportion, while the Gini index and the cross-entropy use all of the class proportions. They respond to improvements in purity that leave the error rate unchanged.
- II is false: such a split leaves the classification error rate unchanged, but it can increase node purity. A purer node gives more certainty about the predicted class of an observation that falls in it.
- III is true: $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$, and each term is close to 0 when $\hat{p}_k$ is close to 0 or 1. A small Gini index means the node is nearly pure.

Problem 10.4. Determine which of the following statements concerning classification trees is/are true.

I. The predicted class for an observation is the most commonly occurring class among the training observations in the terminal node it belongs to.
II. The classification error rate is generally preferred to the Gini index as the criterion for growing the tree.
III. When pruning the tree, the classification error rate is preferable if prediction accuracy of the final pruned tree is the goal.

a. I only
b. II only
c. I and II only
d. I and III only
e. I, II, and III

Solution. The correct answer is **d.**

- I is true: each terminal node predicts its majority class.
- II is false: the classification error rate is not sensitive enough for growing the tree, so the Gini index or the cross-entropy is generally used instead.
- III is true: any of the three measures can be used for pruning, but the classification error rate is preferable when the goal is prediction accuracy of the final pruned tree.